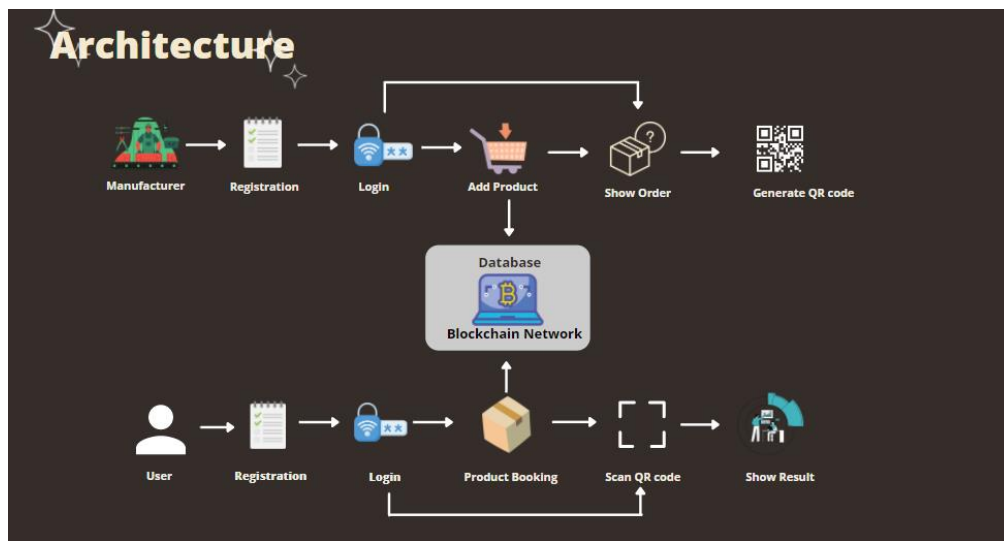


Architecture:



We proposed a system using blockchain technology as an android application for the tracing of products back to its source and for detection of counterfeit products. The proposed system ensures that the products bought are safe and sustainable right from its source which are used in day-to-day life. The proposed system consists of three main parts, customer or user android application, Manufacturer's or company's android application, and Cloud/ Database.

The first application is the Manufacturers or company side application in which we have to first register ourselves. After registration login into the application, we have some options. One option is to add a product in which the manufacturer can add the product details. Another option is to show the order in which they can see customers' order details and after that, they can decide to accept or reject the order. The manufacturer also can see if the product is delivered or not.

A second application is the Customer application in which we have to first register in-app after that we can log in to the application using id and password. In this application, there is an option to show products where customers can see the product details like name, total quantity, price of a product, details of the manufacturer. In that, we can product book the product by inserting the quantity of product. In this application, we can see orders using show my order where we can see the product details, name, quantity, date, time, price, and status of product which is produced is delivered or not. In this app we have a QR code scanner in which we scan the QR code of the product then it shows the source of the product. There is another option which is a blockchain in which it displays the name of generated block product quantity, generated Hash Value, and if the product is corrupted or not.

In this project there is a customer login, in the application. After login, he fills in the details for ordering the product and book the product. The order of the product can be shown to the manufacturer. Manufacturer deciding whether the product request is acceptable or not. After the manufacturer accepting the order of the product it generates the unique QR code of a product. Once an order of product is stored on the network hash code is generated of that product and it is possible to maintain the transaction of the product. In the proposed system QR code is generated for a particular product. Customers scan the QR code on the product or package using the smartphone's QR code reader application or customer application have the option of QR code scanner. After scanning, we get the result of the product source. In the end, the Blockchain system holds these product details along with a history of transactions to enable the tracking of the product along the distribution chain. All the product details, block name, a hash value is stored in the firebase cloud database.